

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Power Electronics	Course Code:	EE3032
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	27-February 2025	Total Marks:	50
	Section:		Weight:	4
	Exam Type:	BEE-6A	Page(s):	1
		Assignment- 3 (Problem Based Learning)	CLO #	3

Student Name: _____ **Roll No.** _____ **Section:** _____

- Instructions:**
1. Use A4-sized sheets for this assignment.
 2. Avoid cutting and overwriting.
 3. Secure the pages with staple, clip, or binding. Do not submit loose pages.
 4. Assignment must be submitted on or before deadline at google classroom.
 5. Late submission will incur a penalty of 20% per day. After 2 days, assignment will no longer be accepted.

Problem Statement:

In high-speed power electronics applications, MOSFETs are widely used for switching operations due to their efficiency and fast response. However, the choice of the gate driver significantly impacts the performance of the circuit. A 7406 open-collector inverter is often considered for MOSFET switching, but it has several limitations in driving high-speed MOSFETs, especially in high-side configurations.

Question No. 1 (CLO No. 3) **Marks: 15**

What might happen if you design a high side MOSFET switch with 7406 drive to work at more than 30 kHz? Can you elaborate on the reason?

Question No. 2 (CLO No. 3) **Marks: 20**

Design an improved gate drive circuit or propose an alternate solution to solve the problem highlighted in Question 1.

Question No. 3 (CLO No. 3) **Marks: 15**

Justify your proposed solution while comparing it with 7406 drive in terms of switching speed, power dissipation and efficiency.